

Day 2 AM: Live demos

Day 2 AM – Agenda

- Make sure we can all generate a wordcloud from yesterday
- Building topic models
 - Discussion: the opportunities and dangers of topic modelling
- Hierarchical speech clustering
- Comparison wordclouds
 - Discussion: What questions are we trying to ask that this wordcloud answers? Is a wordcloud appropriate?
- Lexical dispersion plots
 - Discussion: How Brian has used them (and advanced versions of them) in research

Brian's Research

- "Natural Language Processing"
 - Lexical Dispersion Plots looking at RWE content online
 - (There is a strategic overlap of screenshots there, major content warnings if you look at the reports)
 - <https://zenodo.org/record/4071472> and <https://zenodo.org/record/5732611>
 - Topic Modelling: "latent dirichlet allocation" for MRes thesis supervision

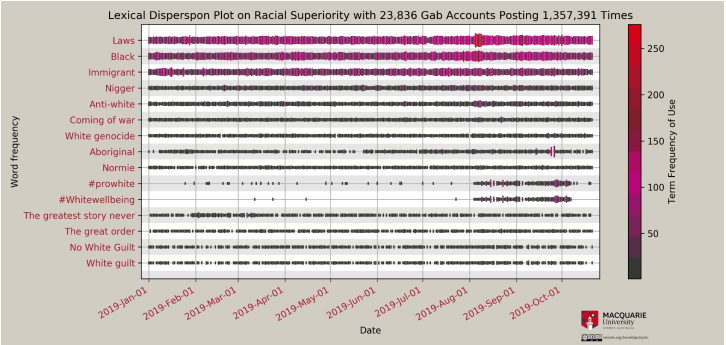


Table 1: Topics from the whole magazine corpus

Topic 1	Topic 2	Topic 3	Topic 4	Topic 5	Topic 6	Topic 7	Topic 8
cup	minute	sugar	min	skin	good	fruit	meat
tomato	heat	cup	chicken	hair	food	place	pork
pasta	pan	place	heat	colour	fresh	fresh	fresh
cheese	cup	min	cup	good	recipe	cup	coupon
min	place	cream	oil	body	new	top	use
sauce	oil	mixture	sauce	new	time	salad	cooking
oil	potato	butter	onion	time	day	slice	clean
top	egg	chocolate	rice	day	store	cheese	time
heat	butter	bowl	bowl	natural	great	bowl	day
garlic	bowl	egg	coriander	help	year	apple	product
parmesan	onion	pan	minute	vitamin	family	salmon	good
bread	lamb	top	place	healthy	product	avocado	valid
basil	garlic	minute	large	food	available	sugar	flavour
ham	large	heat	garlic	use	kid	water	store
olive	mixture	cake	top	product	delicious	flavour	turkey
pan	water	cool	leaf	water	easy	great	great
minute	salt	flour	salad	dry	meal	sweet	chicken
spinach	pepper	smooth	pan	available	range	juice	perfect
salad	season	vanilla	cook	easy	australia	berry	free

Brian: “AI” (Large Language Models) for data analysis

- BERTopic
- Sentiment analysis with LLM
- Generative AI “summarisation”

Before we end, sticky note feedback

- On the green sticky note, write one thing that went well this session.
- On the red sticky note, write one thing that we can improve on, tomorrow.

